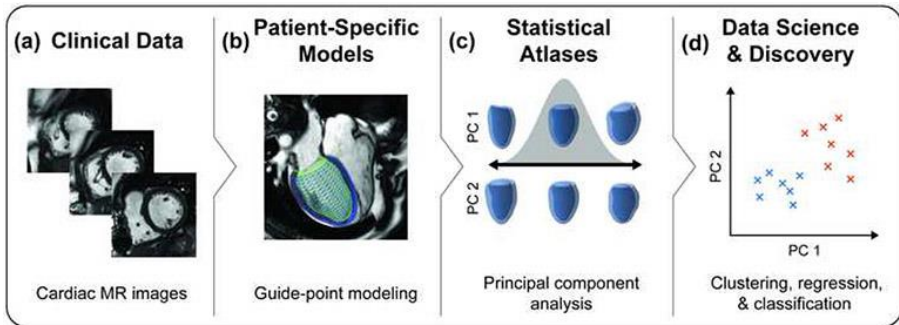


Statistical Shape Models from Clinical Cardiac MR Exams



CAP
Cardiac Atlas Project

Forsch N, et al. *J Comp Sci* 2021



Alistair Young



Sachin Govil



Lisa Parkewitz



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Breandan Crabb



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Albert Hsiao



Marica Muffaletto

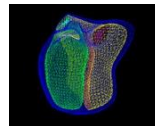
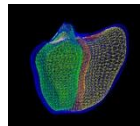
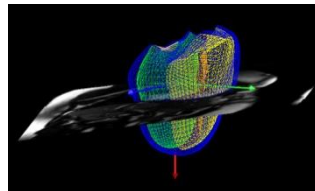
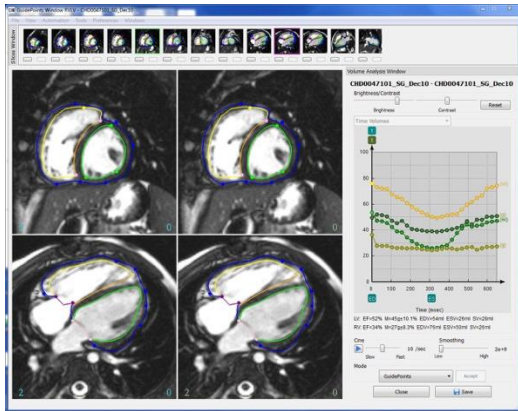


Anna Qi



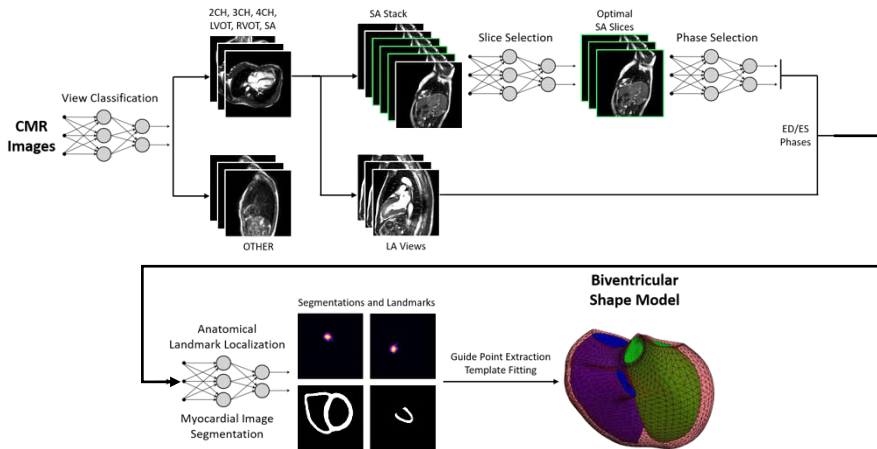
Sanjeet Hegde

Guide-point modeling for biventricular shape model generation



Cardiac Image Modeller (CIM); Auckland, New Zealand

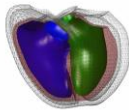
Deep Learning Pipeline to Automate Atlas Generation



Govil S, Crabb BT, Deng Y, Toso LD, Puyol-Anton E, Pushparajah K, Hegde S, Pery JC, Omens JH, Hsiao A, Young AA, McCulloch AD (2023) A Deep Learning Approach for Fully Automated Cardiac Shape Modeling in Tetralogy of Fallot. *Journal of Cardiovascular Magnetic Resonance*.

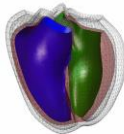
Biventricular Shape Modes in Tetralogy of Fallot

Mode 1



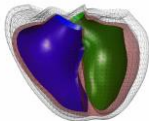
Overall size

Mode 2



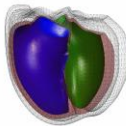
Septal-free wall dimension

Mode 3



Basal vs apical RV bulging

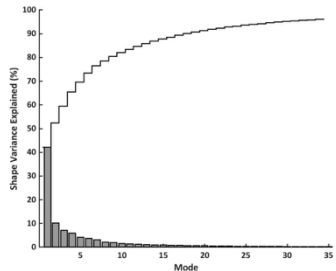
Mode 4



Dilation of RV outflow tract & RV apex

First 30 ED Shape Modes

Explain >95% of the shape variation in the population



Wireframe shows model at ED and shaded surfaces show the model at ES

Govil S, Mauger CA, Hegde S, Occleshaw CJ, Yu X, Perry JC, Young AA, Omens JH, McCulloch AD (2023) Biventricular Shape Modes Discriminate Pulmonary Valve Replacement in Tetralogy of Fallot Better Than Imaging Indices. *Scientific Reports*

Shape PCs stratify future pulmonary replacement better than conventional MR indices

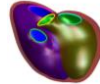
ED4



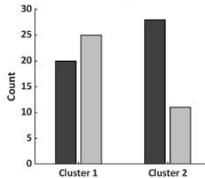
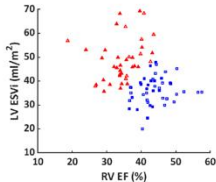
ED6



ED25



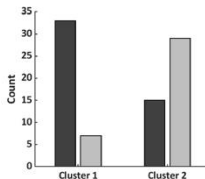
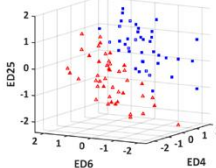
Imaging Indices



MCC = 0.28

Predicted Class	Actual Class	
	PVR	No-PVR
PVR	28	11
No-PVR	20	25

Shape Modes



84 repaired TOF patients:
48 with subsequent
pulmonary valve
replacement

MCC = 0.49

Predicted Class	Actual Class	
	PVR	No-PVR
PVR	33	7
No-PVR	15	29

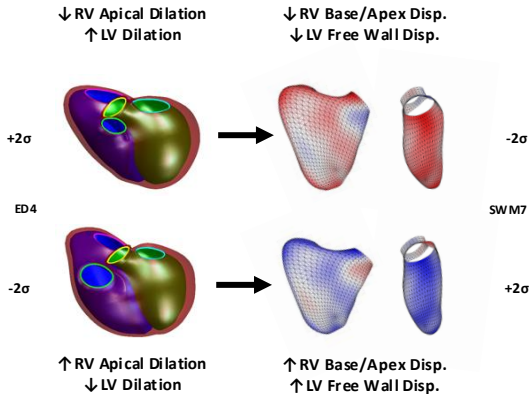
Govil et al. 2023

Govil S, Mauger CA, Hegde S, Occleshaw CJ, Yu X, Perry JC, Young AA, Omens JH, McCulloch AD (2023) Biventricular Shape Modes Discriminate Pulmonary Valve Replacement in Tetralogy of Fallot Better Than Imaging Indices. *Scientific Reports*

Significant shape PCs correlate with systolic wall motion PCs having greatest effect on LV & RV function

Pearson r

ED Mode	1	2	4	7	8
4	0.1	-0.15	0.12	-0.32**	0.25
6	-0.12	0.08	0.16	-0.24	-0.29**
25	-0.03	-0.17	0.01	-0.04	-0.21
	1	2	4	7	8
	SWM Mode				



Govil S, Mauger CA, Hegde S, Occleshaw CJ, Yu X, Perry JC, Young AA, Omens JH, McCulloch AD (2023) Biventricular Shape Modes Discriminate Pulmonary Valve Replacement in Tetralogy of Fallot Better Than Imaging Indices. *Scientific Reports*

Principal Component Analysis (PCA)

See Notebook Example 26.01 in the Book

- ▶ PCA is a dimensionality reduction technique that captures the primary modes of variation in data.
- ▶ PCA finds a set of orthogonal bases or axes (principal components) that are linear combinations of the original features that maximize variance across the observations.
- ▶ Useful for compressing or visualizing high-dimensional data.
- ▶ PCs are ordered by the amount of variance they explain from greatest to least.
- ▶ Given a dataset with n observations (samples) and p features (measurements), put the data in a $n \times p$ matrix $[X]$, where:
 - ▶ Each row is a data sample.
 - ▶ Each column is a feature centered around the mean μ of each column: $\bar{X}_i = X_i - \mu$

PCA uses the Singular Value Decomposition (SVD)

- ▶ Compute the Singular Value Decomposition (SVD):
 $[X] = [U][\Sigma][V]^T$
- ▶ The covariance matrix is:
 $[C] = \frac{1}{n-1}[\bar{X}]^T[\bar{X}] = [V]\left(\frac{[\Sigma]^2}{n-1}\right)[V]^T$
- ▶ The columns of $[V]$ are the eigenvectors of $[C]$ (principal directions)
- ▶ $[\Sigma]$ is the diagonal matrix of singular values,
 $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_p \geq 0$, where $\sigma_i^2/(n-1)$ is the variance explained by the i -th PC
- ▶ The columns of $[U]$ are the PC scores (how each sample projects onto each PC).
- ▶ Truncating $[\Sigma]$ and $[V]$ and projecting the original data onto the new axes: $[Z] = [X][V]$ gives the coordinates of the data in the reduced PC space.

Tools for Ventricular Statistical Shape Modeling

See Notebook Example 26.1 in the Book



```
from sklearn.decomposition import PCA
pca = PCA(n_components=10)
pca_data = pca.fit_transform(measurements)
explained_variance = pca.explained_variance_ratio_
```



Cardiac Atlas Project

<https://www.cardiacatlas.org/biventricular-modes/>

SSSCP Book: 26.1_example_explore_atlas.ipynb

[https://sscp.simula.no/user/admcculloch/lab/workspaces/autot/tree/SSCP_lectures/L10%20\(Cardiac%20Mechanics\)/book/26.1_example_explore_atlas.ipynb](https://sscp.simula.no/user/admcculloch/lab/workspaces/autot/tree/SSCP_lectures/L10%20(Cardiac%20Mechanics)/book/26.1_example_explore_atlas.ipynb)

This notebook uses the h5py Python library that provides an interface to an HDF5 (Hierarchical Data Format 5) file with the data classes defining a biventricular ED/ES atlas derived from UK Biobank imaging substudy CMR Exams: UKBRVLV_All.h5